

CAPSTONE PROJECT 2

SIMPLE SYSTOLIC ARRAY MULTIPLIER (8-BIT INTEGER)

Design. Verify. Optimize.

1. PROJECT OVERVIEW

Design and implement an **8-bit Integer Systolic Array Multiplier** using Verilog HDL. The array will compute the product of two 8-bit unsigned integers (A and B) and generate a 16-bit result. The design should demonstrate the parallelism and regular structure of systolic arrays.

2. OBJECTIVES

- Understand systolic array architecture
- Implement a regular 2D array of Processing Elements (PEs)
- Perform multiply-accumulate operations in a systolic manner
- Verify the design with a self-checking testbench
- Analyze performance (latency and throughput)

3. DESIGN SPECIFICATIONS

Parameter	Specification
Operand Width	8-bit Unsigned (A[7:0], B[7:0])
Output Width	16-bit Unsigned (P[15:0])
Architecture	2D Systolic Array Multiplier
Array Size	8 x 8 Processing Elements (PEs)
Operation	Multiply-Accumulate ($M \times A + C$)
Data Flow	Systolic (Wavefront Propagation)
Clock Frequency (Target)	50 MHz (for simulation)
Language	Verilog HDL
Reset	Active Low Synchronous
Throughput	1 result per clock cycle (after pipeline fill)
Latency	(2N-1) cycles = 15 cycles for N=8

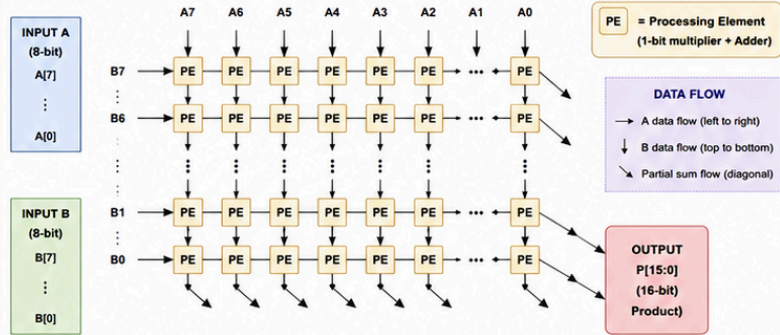
9. TESTBENCH REQUIREMENTS

- Self-checking testbench
- Apply at least 100 random test vectors
- Directed test cases (edge cases):
 - 0×0 , $0 \times \text{MAX}$, $\text{MAX} \times 0$, $\text{MAX} \times \text{MAX}$
 - Power of 2, alternating patterns
- Compare output with expected result ($A \times B$)
- Display PASS/FAIL summary
- Generate waveform

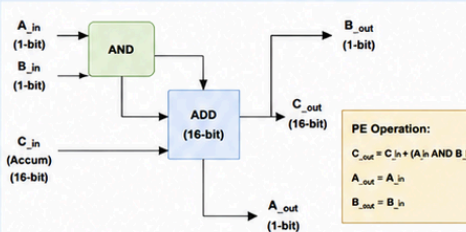
13. GENERAL GUIDELINES

- Follow good coding practices and meaningful naming conventions.
- Use parameterized code where applicable.
- Ensure synthesizable RTL and clean simulation.
- No use of built-in $**$ operator for full 8x8 multiplication.
- Only structural/behavioral description using PE array.

4. BLOCK DIAGRAM



5. PROCESSING ELEMENT (PE) ARCHITECTURE



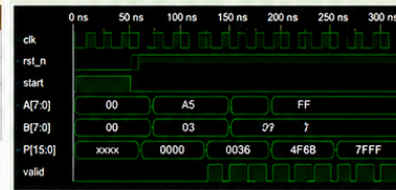
6. SYSTOLIC OPERATION

- A bits flow horizontally from left to right.
 - B bits flow vertically from top to bottom.
 - Partial sums flow diagonally from top-left to bottom-right.
 - After initial latency (2N-1) cycles, the array produces one result per clock cycle.
- For N = 8 $\rightarrow$ Latency = 15 cycles.

7. TIMING / PIPELINE BEHAVIOR

Phase	Description
1. Data Injection	A and B bits are injected into the array (MSB first).
2. Propagation	Data moves through the PEs in systolic manner.
3. Output Generation	Partial sums reach the last row; final sum available at P[15:0].
4. Steady State	One result per clock cycle.

8. SIMULATION WAVEFORM (SAMPLE)



Note: Valid goes high when P is ready. After 15 cycles, one result per clock.

10. DELIVERABLES

- Verilog RTL Code (design.sv)
- Testbench (tb_systolic_multiplier.sv)
- Simulation Waveform (screenshot)
- Synthesis Report (optional)
- Project Report (PDF)
- Design Documentation (architecture + flow)

11. EVALUATION CRITERIA

Criteria	Weightage
Functionality (Correctness)	40%
RTL Design Quality	20%
Testbench Coverage	20%
Waveform & Results	10%
Documentation	5%
Timely Submission	5%
Total	100%

12. SUBMISSION DETAILS

- Repository (GitHub Classroom): <https://classroom.github.com/a/cxxxxxx>
- Branch Name: main
- Commit Message: "Capstone2_Systolic_FINAL"
- Submission Date: **7 DAYS** from Project Release Date (by 11:59 PM IST)

Build it. Verify it. Optimize it. 🚀